

HE2003 Econometrics I

Tutorial 11

11/11/2024

Question 1

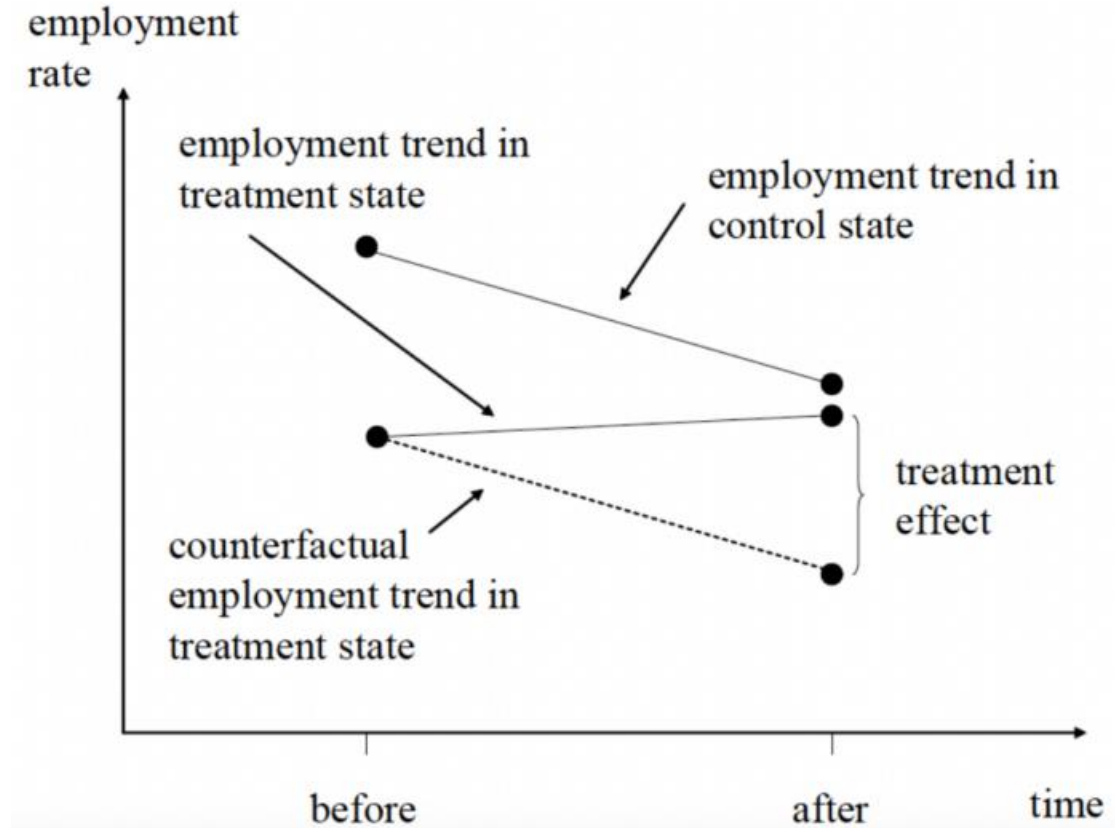
- a) Explain Card and Krueger's seminal study regarding minimum wage and employment by using the difference-in-differences (DID).

This is one of the first applications of difference-in-differences (DID).

- New Jersey (NJ) raised the state minimum wage from \$4.25 to \$5.05 in April, 1992. The minimum wage in the neighboring state Pennsylvania (PA) stayed at \$4.25.
- Card and Krueger collected data on employment at fast food restaurants in NJ in Feb 1992 (before the policy change) and again in Nov 1992 (after the policy change). They collected data from the same type of restaurants in eastern PA.
- They compared the Feb-to-Nov change in employment in NJ to the change in employment in PA over the same period to obtain the DID estimate of the effect of minimum wage change on employment.

Question 1

See the figure for the graphical illustration. Note that to obtain the treatment effect, we need to assume that without the policy change (the raise in minimum wage), the NJ trend (treatment trend) would have moved in the same way as the PA trend (control trend). This is the key assumption of the DID design, which is called the **“parallel trend assumption”**.



Question 1

Then, the DID estimate of the policy effect is:

$$\begin{aligned} DID &= [\bar{Y}_{After,NJ} - \bar{Y}_{before,NJ}] - [\bar{Y}_{After,PA} - \bar{Y}_{before,PA}] \\ &= (20.90 - 20.43) - (21.10 - 23.38) = 2.75 \end{aligned}$$

where $\bar{Y}_{After,NJ}$ denotes the average employment (in fast-food restaurants) in NJ after the policy change, etc. Therefore, Card and Krueger's DID analysis shows that the employment in NJ was not negatively affected by the raise of the minimum wage. Their empirical result is in contrast to the prediction from the standard economic theory.

	New Jersey	Pennsylvania	Difference
Before Increase	20.43	23.38	2.95
After Increase	20.90	21.10	0.20
Difference	0.47	-2.28	2.75

Average employment at fast food restaurants in NJ and PA

Question 1

b) Explain the linear-log model.

The linear-log regression model (with a single regressor) can be written as:

$$Y_i = \beta_0 + \beta_1 \log(X_i) + u_i, \quad i = 1, \dots, n$$

where $\log(X_i)$ is the natural logarithm of X_i . The only difference between this model and the standard linear regression model is that the RHS variable is now $\log(X_i)$ rather than X_i itself.

The meaning of β_1 is as follows:

$$\beta_1 \approx \frac{\Delta Y}{\Delta X/X} = \frac{\text{change in } Y}{\text{percentage change in } X}$$

For example, suppose $X = 100$ and $\Delta X = 1$, then $\Delta X/X = 1\%$. Suppose $\beta_1 = 2$, then the corresponding change in Y is: $\Delta Y \approx \beta_1 \times \Delta X/X = 2 \times 1\% = 0.02$ (unit of Y). Similarly, suppose $\Delta X/X = 10\%$ and $\beta_1 = 2$, then the corresponding change in Y is: $\Delta Y \approx \beta_1 \times \Delta X/X = 2 \times 10\% = 0.2$ (unit of Y).

Question 2

Suppose that in 2021 a city with two flagship universities was given a sharp increase in funding. Call the two universities Eastern and Western, respectively. The local government decided that the funding would go to Western, mostly to hire more teachers to reduce class sizes. To evaluate the effectiveness of the increased funding, the government collected random samples of the students in 2020 and 2022 at both universities. The measure of student performance is a standardized test score given to all undergraduate students in the city.

- a) Let *score* be the test score. Without controlling for other factors, write down a difference-in-differences (DID) model using a linear regression with an interaction term that allows you to determine whether the additional funding led to improved test scores. Define precisely all the regressors (explanatory variables) in your model. Explain the specific meaning of all the parameters in the model.
- b) Illustrate your DID model by using a diagram. What is the key assumption for your model? Explain.
- c) How might estimating the model in (a) be biased if Western gets an influx of better students because of the increased funding? Explain.

Question 2 (a)

The difference-in-differences (DID) model can be defined as follows:

$$score = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 D_1 \times D_2 + u$$

where $D_1 = 1$ for Western (treatment group), $D_1 = 0$ for Eastern (control group), $D_2 = 1$ for the time period after the treatment (2022), and $D_2 = 0$ for the time period before the treatment (2020). Then, we have totally four scenarios:

$$\{D_1 = 0, D_2 = 0\} = \{Eastern, Before Treatment (fund increase)\}$$

$$\{D_1 = 1, D_2 = 0\} = \{Western, Before Treatment (fund increase)\}$$

$$\{D_1 = 0, D_2 = 1\} = \{Eastern, After Treatment (fund increase)\}$$

$$\{D_1 = 1, D_2 = 1\} = \{Western, After Treatment (fund increase)\}$$

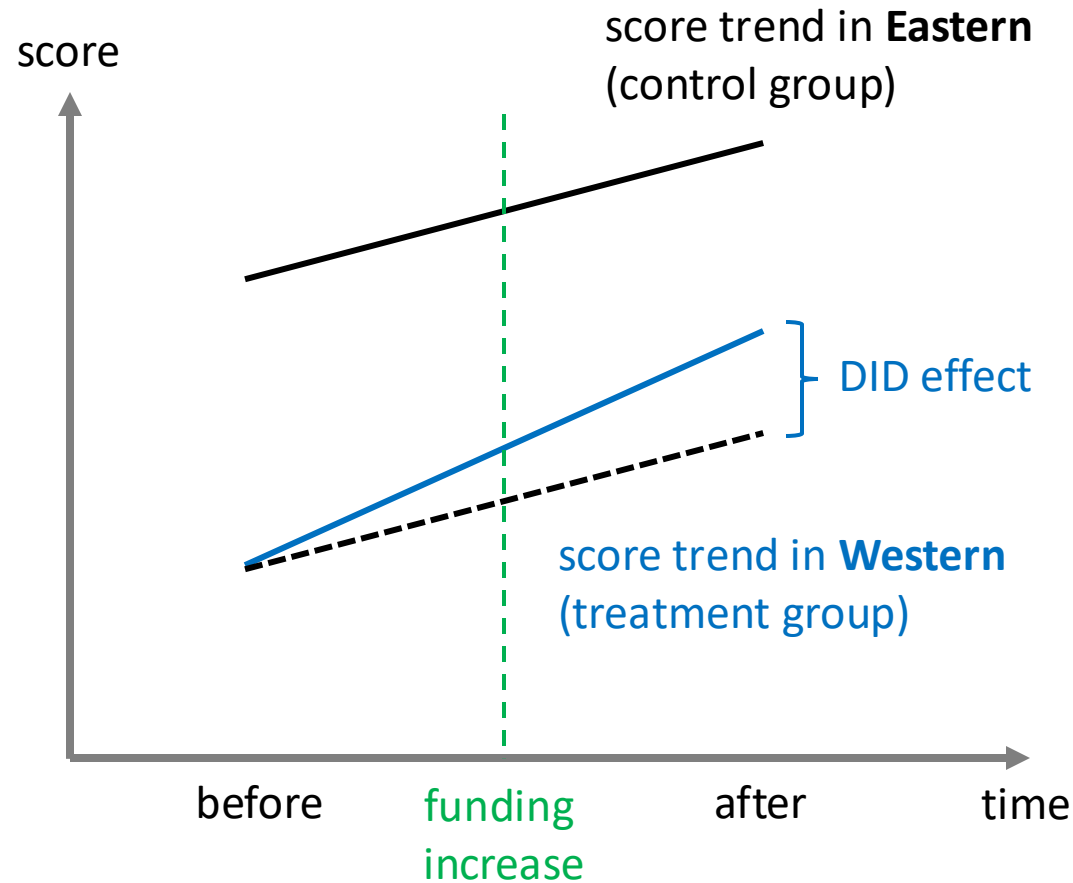
Question 2 (a)

It is also clear that β_3 (the parameter for the **interaction between the two dummy variables**) corresponds to the DID treatment effect:

$$\begin{aligned}\beta_3 &= [\{Western, After Treatment\} - \{Eastern, After Treatment\}] \\ &\quad - [\{Western, Before Treatment\} - \{Eastern, Before Treatment\}] \\ &= [\{Western, After Treatment\} - \{Western, Before Treatment\}] \\ &\quad - [\{Eastern, After Treatment\} - \{Eastern, Before Treatment\}]\end{aligned}$$

Therefore, β_3 is the key parameter of interest in the model: Whether the additional funding for Western led to improved test scores is captured by β_3 .

Question 2 (b)



Parallel trend assumption: Without the policy change (funding increase), the Western trend (treatment trend) would have moved in the same way as the Eastern trend (control trend).

Black dashed line is **counterfactual** score trend for Western

In the figure, we suppose the Eastern trend is higher than the Western trend.

Question 2 (c)

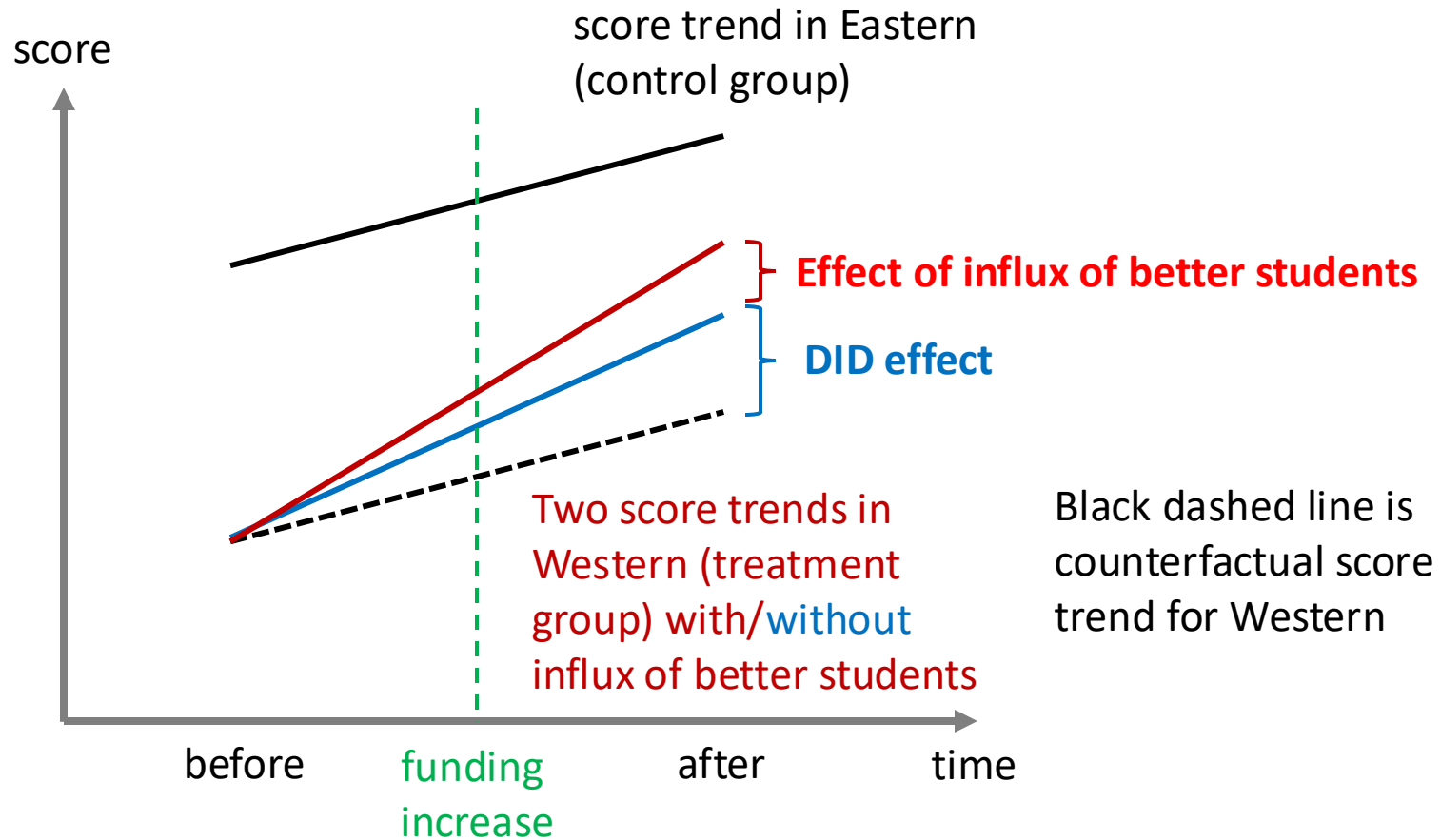
The influx of better students to Western may induce an upward bias for the DID estimator. This can also be seen as follows:

$$DID\ estimator = (\overline{score}_{W,After} - \overline{score}_{W,Before}) - (\overline{score}_{E,After} - \overline{score}_{E,Before})$$

where $\overline{score}_{W,After}$ denote the average score of Western after the treatment (additional funding), etc. Then, we expect that $\overline{score}_{W,After}$ will increase due to both the fund increase and better students, so that the value of the DID estimator will be higher than it should be. Intuitively, there are **now two factors driving the DID estimator**: (a) **the effect of the fund increase for Western** (this is the treatment effect we are really interested in), and (b) **the effect of the better students to Western**. This results in an **upward bias** for the DID estimator.

The graphical illustration is shown in the next figure.

Question 2 (c)



Question 3

Consider the following two specifications of regression models:

$$Y = \beta_0 + \beta_1 Treat + \beta_2 Treat \times After + u \quad (1)$$

$$Y = \beta_0 + \beta_1 After + \beta_2 Treat \times After + u \quad (2)$$

where *Treat* is a dummy variable that equals 1 for Treatment Group and equals 0 otherwise; *After* is a dummy variable that equals 1 for the time period after the treatment/policy change and equals 0 otherwise.

- Explain the specific meaning of the parameters $\beta_0, \beta_1, \beta_2$ in the model (1). Illustrate the model by using a diagram.
- Explain the specific meaning of the parameters $\beta_0, \beta_1, \beta_2$ in the model (2). Illustrate the model by using a diagram.
- Compare the two above models with a difference-in-differences model. Which model is better? Explain.

Question 3 (a)

For the model $Y = \beta_0 + \beta_1 Treat + \beta_2 Treat \times After + u$, we have:

$$E[Y|Treat = 0, After = 0/1] = \beta_0$$

$$E[Y|Treat = 1, After = 0] = \beta_0 + \beta_1$$

$$E[Y|Treat = 1, After = 1] = \beta_0 + \beta_1 + \beta_2$$

Therefore, β_0 is the average outcome of the control group (for both before and after). **In a graphical illustration, the trend of the control group should be totally horizontal.**

In addition, we have

$$\beta_1 = E[Y|Treat = 1, After = 0] - E[Y|Treat = 0, After = 0/1]$$

which is the difference in average outcomes between the treatment group before treatment and the control group; and

$$\beta_2 = E[Y|Treat = 1, After = 1] - E[Y|Treat = 1, After = 0]$$

which is the difference in average outcomes of the treatment group after and before the treatment

Question 3 (b)

Similarly, for the model $Y = \beta_0 + \beta_1 After + \beta_2 Treat \times After + u$, we have:

$$E[Y|Treat = 0/1, After = 0] = \beta_0$$

$$E[Y|Treat = 0, After = 1] = \beta_0 + \beta_1$$

$$E[Y|Treat = 1, After = 1] = \beta_0 + \beta_1 + \beta_2$$

Therefore, β_0 average outcome of both treatment and control groups before the treatment. **In the graph, the trends of both treatment and control groups should start from the same point.**

Furthermore, we have

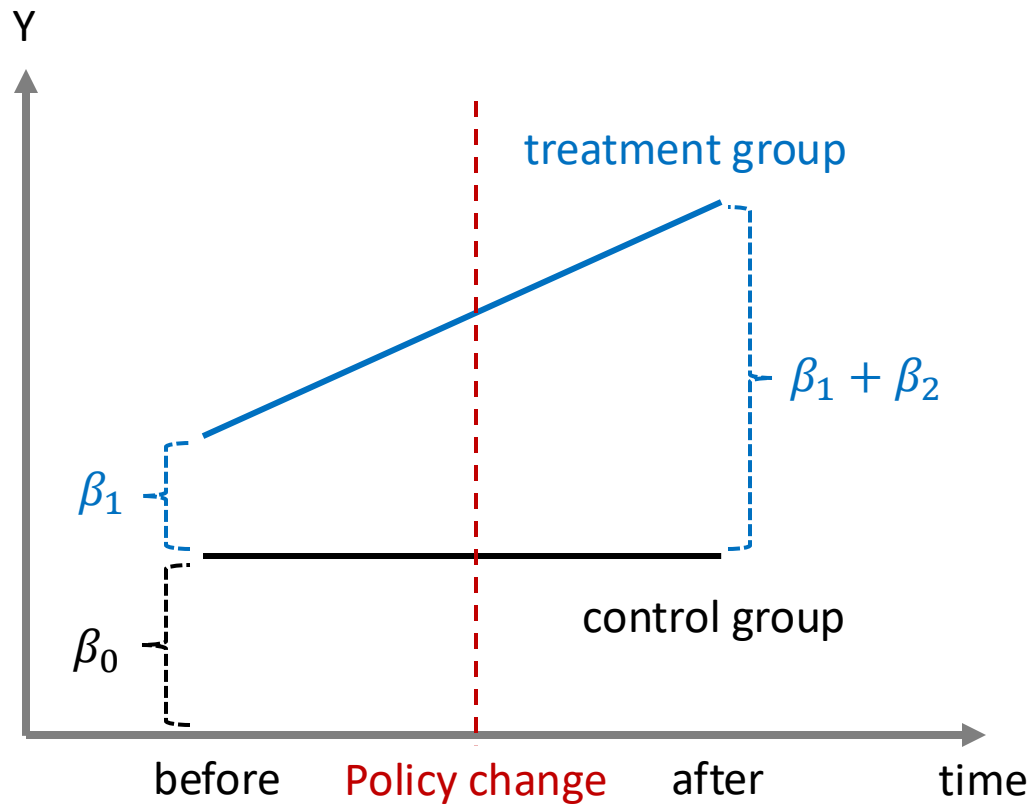
$$\beta_1 = E[Y|Treat = 0, After = 1] - E[Y|Treat = 0/1, After = 0]$$

which is the difference in average outcomes between the control group after the treatment and the treatment/control groups before treatment; and

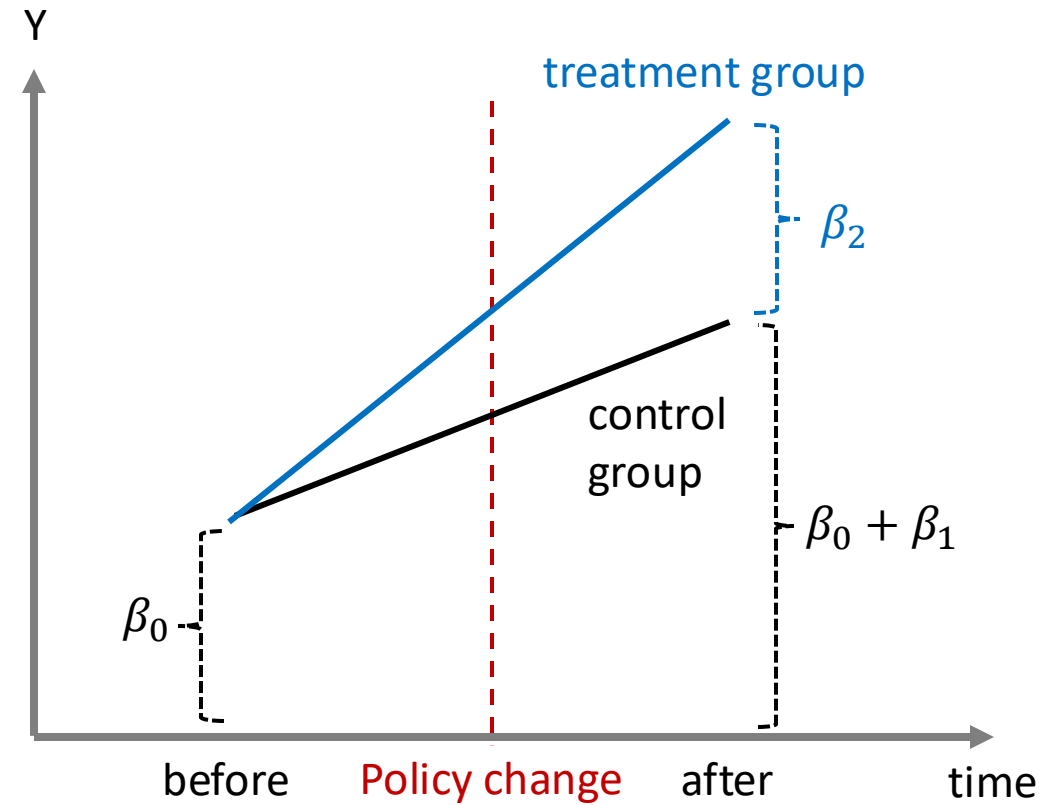
$$\beta_2 = E[Y|Treat = 1, After = 1] - E[Y|Treat = 0, After = 1]$$

which is the difference in average outcomes between the treatment and control groups after the treatment

Question 3 (a) & (b)



(a)



(b)

Question 3 (c)

For a correct difference-in-differences (DID) model, the meaning of the coefficient of the interaction term is:

$$(E[Y|Treat = 1, After = 1] - E[Y|Treat = 1, After = 0]) \\ - (E[Y|Treat = 0, After = 1] - E[Y|Treat = 0, After = 0])$$

which is equal to “difference of the treatment group before and after the treatment” – “difference of the control group before and after the treatment” (note that this is equivalent to “difference between two groups after the treatment” – “difference between two groups before the treatment”).

Therefore, neither of the specifications in (a) and (b) is correct to obtain the parameter of interest. In particular, the model in (a) does not allow for the change in outcome of the control group before and after the treatment, while the model in (b) does not allow for the difference in outcomes between the two groups before the treatment. All the dummy variables $Treat$, $After$, and $Treat \times After$ are necessary for the correct specification of DID model.

Question 4

Acemoglu, Johnson, and Robinson (AJR, 2024 Nobel Laureates in Economics) are interested in the effect of good political institutions (e.g., strong rule-of-law, protection of property rights) on prosperity and long-term economic performance. They used a sample of 64 former European colonies and argue that European colonizing countries tended to set up colonies as either an “extractive state” or as a “migrant colony”.

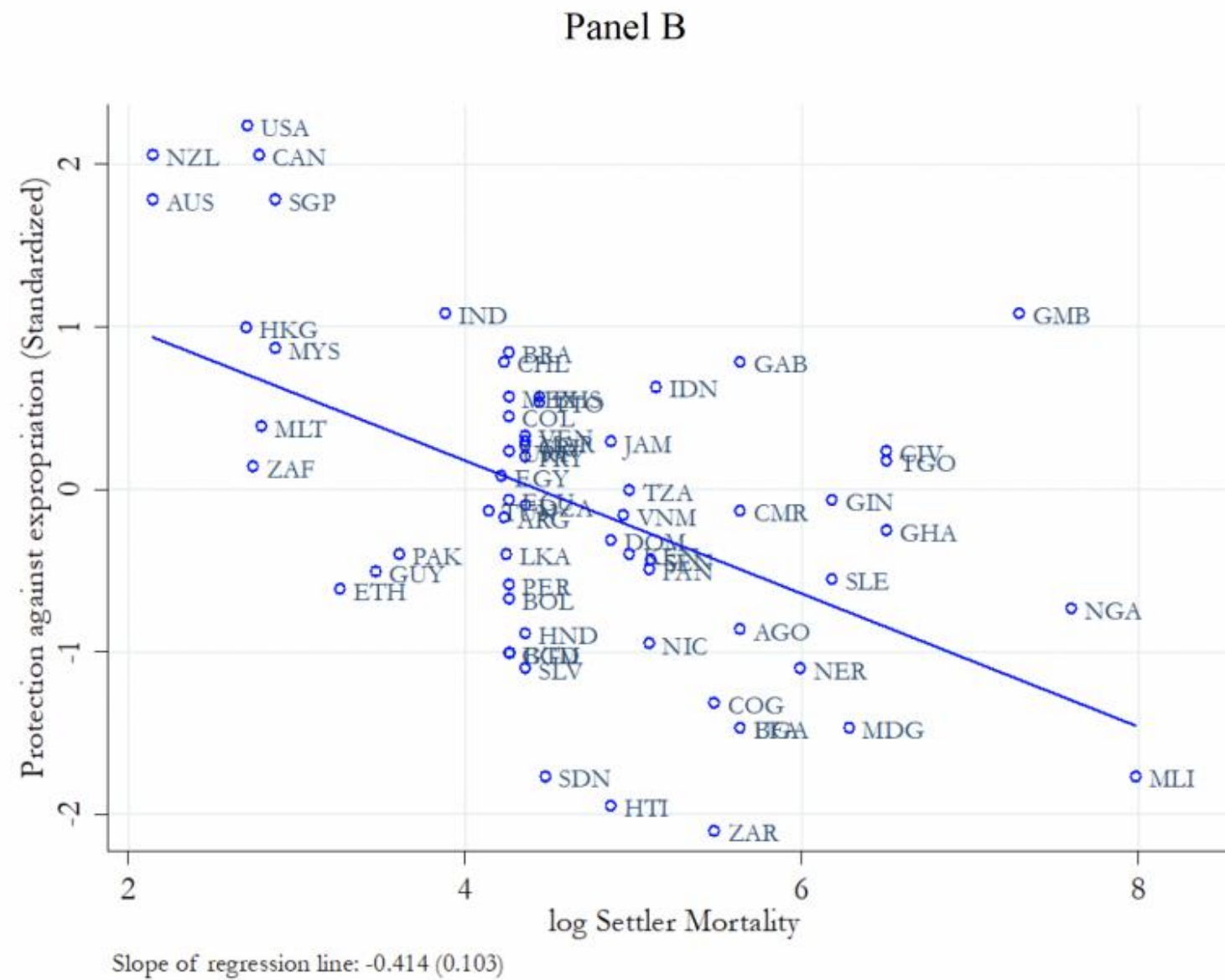
AJR highlighted that initial conditions in areas colonized by Europeans differed largely by disease environments, in particular for the settlers themselves. They hypothesized that Europeans settled to a greater extent in places where their mortality rates were relatively low; they brought along their values as well as beliefs and developed home-like institutions, in particular economic and political institutions benefitting themselves. In areas where mortality was high due to diseases such as malaria and yellow fever, Europeans did not settle in large numbers. In these places, therefore, the colonizers instead established or maintained institutions designed to extract resources from native populations as much and as quickly as possible. Many characteristics of the colonial state and institutions persisted even after independence, and that such persistence influences contemporary economic performance.

Question 4

- a) Discuss which countries in Panel B are more likely to be “extractive state” and “migrant colony”, respectively.
- b) Construct a linear-log model for “Protection Against Expropriation Risk” (1985-95) and “Settler Mortality” (19th century) and explain the meaning of the slope estimate -0.414 .

Figure 2: Proposed causal chain in Acemoglu, Johnson, and Robinson (2001)

1. Living conditions during colonization (settler mortality) $\Rightarrow$ 2. Size of European settlements during colonization $\Rightarrow$ 3. Colonial institutions $\Rightarrow$ 4. Institutions at independence $\Rightarrow$ 5. Contemporary institutions $\Rightarrow$ 6. Contemporary economic prosperity



Notes. Panel A: Relationship between log GDP per capita in 1995 (purchasing power adjusted) and log of settler mortality and. Panel B: Relationship between average protection against expropriation risk 1985-95 and log of settler mortality. Panels A and B are analogous to Figures 1 and 3 in Acemoglu, Johnson, and Robinson (2001). They have been reproduced using data from Acemoglu, Johnson, and Robinson (2012).

Question 4 (a)

According to AJR's argument, the colonies where the early European settlers' mortality rates had been low were more likely to become the "migrant colonies" (some scholars also called them "Neo-Europe"). By contrast, the colonies where the mortality rate had been high were more likely to become the "extractive states".

Therefore, in Panel B, the left-hand-side countries such as "NZL", "AUS", "USA", "CAN", and "SGP" are more likely to be the "migrant colonies", while the righthand-side countries such as "MLI", "NGA", "MDG", and "GHA" are more likely to be the "extractive states".

Question 4 (a)

The linear-log model can be written as follows:

$$Protection_i = \beta_0 + \beta_1 \log(Mortality_i) + u_i, \quad i = 1, \dots, n$$

where $Protection_i$ denotes the index of “Protection Against Expropriation Risk” (1985-95) for the i -th former European colony, and $Mortality_i$ denotes the “Settler Mortality” (19th century) for the i -th colony. Given the definition of the linear-log model,

$$\beta_1 \approx \frac{\Delta Protection}{\Delta Mortality / Mortality}$$

Therefore, supposing that Mortality changes by 1 percent, we have

$$\Delta \widehat{Protection} \approx \hat{\beta}_1 \times \frac{\Delta Mortality}{Mortality} = -0.414 \times 1\% = -0.00414 \text{ (unit of } Protection \text{)}$$

Question 5

Empirical Exercise: Use the dataset **lawsch ranking.dta** uploaded in NTU Learn. The variable *llibvol* is equal to $\log(\text{libvol})$, the log of number of volumes in library, and *lcost* is equal to $\log(\text{cost})$, the log of law school cost such as tuition fee.

- a) Run the regression of *salary* (the starting salary of law school graduates) on *top10*, *r11_25*, *r26_40*, *r41_60*, *GPA*, *llibvol* and *lcost*. Interpret the estimated coefficients of *llibvol* and *lcost*.
- b) Test the hypothesis that ranking does not matter for the starting salary using both homoskedasticity-only and heteroskedasticity-robust *F*-tests.

Stata commands:

```
reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost  
test top10 r11_25 r26_40 r41_60
```

Question 5 (a)

```
. reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost
```

Source	SS	df	MS	Number of obs	=	136
Model	1.8344e+10	7	2.6206e+09	F(7, 128)	=	159.22
Residual	2.1067e+09	128	16458377	Prob > F	=	0.0000
				R-squared	=	0.8970
				Adj R-squared	=	0.8914
Total	2.0451e+10	135	151485812	Root MSE	=	4056.9

salary	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
top10	27756.33	2218.722	12.51	0.000	23366.21	32146.45
r11_25	21724.47	1621.093	13.40	0.000	18516.86	24932.08
r26_40	11070.08	1440.387	7.69	0.000	8220.027	13920.13
r41_60	6456.605	1174.121	5.50	0.000	4133.406	8779.805
GPA	7382.992	2762.113	2.67	0.008	1917.679	12848.31
llibvol	3284.982	1176.495	2.79	0.006	957.0853	5612.879
lcost	1779.923	1071.077	1.66	0.099	-339.3859	3899.232
_cons	-26686.04	15308.94	-1.74	0.084	-56977.4	3605.31

Question 5 (a)

Define the regression model:

$$\begin{aligned} \text{Salary} = & \beta_0 + \beta_1 \text{top10} + \beta_2 r11_25 + \beta_3 r26_40 + \beta_4 r41_60 \\ & + \beta_5 \text{GPA} + \beta_6 \log(\text{libvol}) + \beta_7 \log(\text{cost}) + u \end{aligned}$$

The estimated OLS coefficients are 3284.98 ($\hat{\beta}_6$) and 1779.92 ($\hat{\beta}_7$) for *llibvol* and *lcost*, respectively. Both have the representation of a linear-log model, that is,

$$\Delta \widehat{\text{Salary}} \approx \hat{\beta}_6 \times \frac{\Delta \text{libvol}}{\text{libvol}} = 3284.96 \times 1\% = 32.85 \text{ (USD)}$$

$$\Delta \widehat{\text{Salary}} \approx \hat{\beta}_7 \times \frac{\Delta \text{cost}}{\text{cost}} = 1779.92 \times 1\% = 17.80 \text{ (USD)}$$

Question 5 (b)

The homoskedasticity-only F-statistic is equal to 53.13. According to the critical value tables, the 5% critical value for the $\chi^2_4/4$ ($F_{4,\infty}$) is equal to 2.37. So, the null hypothesis is rejected at the 5% level. In addition, we can see from the STATA result below, the p-value of the F-test is zero.

```
. test top10 r11_25 r26_40 r41_60
```

```
( 1)  top10 = 0
```

```
( 2)  r11_25 = 0
```

```
( 3)  r26_40 = 0
```

```
( 4)  r41_60 = 0
```

```
      F( 4, 128) =    53.13  
      Prob > F =    0.0000
```

Question 5 (b)

Stata commands with the robust standard error:

```
reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost, robust
```

```
test top10 r11_25 r26_40 r41_60
```

```
. test top10 r11_25 r26_40 r41_60
```

```
( 1) top10 = 0
```

```
( 2) r11_25 = 0
```

```
( 3) r26_40 = 0
```

```
( 4) r41_60 = 0
```

```
F( 4, 128) = 51.37  
Prob > F = 0.0000
```

The heteroskedasticity-robust F-statistic is equal to 51.37, and the 5% critical value is the same as that for the homoskedasticity-only F-test. Therefore, we also reject the null hypothesis at the 5% level.